

How to Avoid Overfitting:

Avoiding overfitting in regression machine learning models shares some similarities with general techniques for avoiding overfitting, but there are also specific strategies tailored to regression tasks. Here are some techniques:

1. **Eda**
2. **For Regression model use Ridge(L2 Regularization)**
3. **Increase Training Data:**
4. **Cross-Validation:**
5. **Feature Selection:**
 - i. **Filter Method**
 - ii. **Wrapper Method**
 - iii. **Embedded Methods**
6. **Feature Extraction (Dimensionality Reduction)**
 - i. **PCA**
 - ii. **LDA**
 - iii. **T-SNE**
7. **Feature Scaling:**
8. **Hyperparameter Tuning:**

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Hyperparameter Tuning:

Hyperparameters are settings or configurations of a machine learning model that are tuned by the practitioner to optimize the model's performance. Here are examples of hyperparameters for various machine-learning models:

Increase Training Data:

If possible, collect more training data to improve the model's ability to generalize. More data can help reduce overfitting by providing the model with a more representative sample of the underlying data distribution.

Cross-Validation:

Split your dataset into training, validation, and test sets. Use techniques like k-fold cross-validation to evaluate your model's performance on multiple splits of the data. This helps ensure that your model's performance isn't overly dependent on a particular subset of the data.

Working with Feature

Feature Selection:

Choose relevant features for your regression model and avoid adding noise or irrelevant features. Techniques like

- **Filter Method:**
- **Wrapper Method:**
- **Embedded Methods**
 - a. **For linear and logistic regression:**
 - i. **Ridge regularization**
 - ii. **Lasso Regularization**
 - b. **Tree like algorithm:**
 - i. **Early Stopping:**
 - ii. **Prepruning**
 - iii. **Post Pruning:**
 - iv. **Ensemble Methods**

Feature Extraction (Dimensionality Reduction) :

It involves transforming raw data into a set of features that can be used for model training and evaluation. There are various techniques for feature extraction, depending on the

type of data you're working with. Here's a rundown of some common methods:

Dimensionality Reduction Techniques

- **Principal Component Analysis (PCA):**
- **Linear Discriminant Analysis (LDA):**
- **t-Distributed Stochastic Neighbor Embedding (t-SNE):**

Feature Scaling:

Normalize or standardize your features to ensure that they are on a similar scale. This can help prevent the model from being overly sensitive to the scale of different features.

